MATHEMATICAL WORKING PAPER

Research as Production Context

Conditional labor requirements, reusable knowledge,
and labor-type conversion ratios

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ABSTRACT

This paper models research, designs, and completed software as part of the context of production. Nonnegative input and occupational labor matrices determine conditional direct and indirect labor requirements. A Boolean closure records reusable capabilities supporting the selected production chain. Research projects use resources and change available capabilities after successful completion. The paper establishes productivity, uniqueness, capability closure, unit invariance, coefficient sensitivity, software specialization, inventory reconciliation, and exact reporting-boundary reduction.

An optional labor-type conversion model adds subsistence, education, training, and productive lifetime data. Under an explicit equal-residual weighting assumption, a second matrix determines positive normalized weights and pairwise ratios. The audited extension proves how these ratios are preserved under exact boundary reduction and correctly transformed under occupational time units. Exact counterexamples identify what cannot be inferred without the weighting assumption. Occupational hours, provisioning burdens, scalar values, and wages remain distinct. The results are conditional mathematical statements, not empirical validation of a unique Marxian reduction of heterogeneous labor.

UNIFIED SPECIFICATION

$$
\begin{aligned}
G_C &= H_C(I - A_C)^{-1}, & R_C &= D_C \odot B_C^*, \\
M_C &= G_C S_C + (G_C E_C + U_C)\Gamma_C^{-1}, & & \\
\beta_C &= (I - M_C^{\mathsf{T}})^{-1}\mathbf{1}, & \alpha_C &= \beta_C/\beta_{C,b}, \\
v_C &= G_C^{\mathsf{T}}\alpha_C, & r_{ij}(C) &= \alpha_{C,i}/\alpha_{C,j}, \\
\ell_p(C) &= h_p + G_C a_p, & V_p(C) &= \alpha_C^{\mathsf{T}}\ell_p(C).
\end{aligned}
$$

The normalized extension assumes $\rho(A_C) < 1$, $\rho(M_C) < 1$, and equal residuals per clock hour. The core occupational model does not require the optional conversion assumption.

Audit result. The core equations and existing numerical examples are unchanged. The revision completes boundary and labor-unit rules, strengthens identification limits, and fixes the portable verification script. All 135 named verification checks pass.

Reader's guide. Original body and references: pages 2–17. Revised conversion appendix: pages 18–25. New proofs, counterexamples, and audit record: pages 26–29. Editable appendix source and verification files are embedded.

Keywords: input-output analysis; labor requirements; research and development; software; reusable knowledge; heterogeneous labor; conditional conversion.

1 Introduction and economic scope

The question addressed here is how to calculate the labor required to reproduce an output when its production depends on reusable research, designs, or software. A product may require a scientific method directly, or it may use a machine whose manufacture requires that method. A software release can similarly enable copying or a service without being used up in proportion to the number of delivered units. The accounting object must specify the production conditions under which reproduction is evaluated.

This paper defines a *conditional labor-requirement model*. Its quantitative part measures direct and indirect labor by occupation. Its logical part records reusable capabilities supporting the selected production chain. Research and development are activities that consume resources and, when successful, change the available capabilities. These activities can therefore alter subsequent production possibilities without imposing a universal rule allocating their historical cost to every future product.

The economic motivation is compatible with making production conditions explicit in a Marxian analysis. Marx relates socially necessary labor to prevailing production conditions and identifies science and its application among the determinants of productivity [1, Ch. 1]. His discussion of machinery distinguishes scientific knowledge from the apparatus through which it is applied [1, Ch. 15, Sec. 2]. These observations motivate the distinction used here; they do not by themselves determine the accounting convention for every form of research.

The quantitative foundation belongs to the input-output tradition associated with Leontief [3]. Boolean matrix closure is also established mathematics; Kozen [4] provides a general algebraic setting. The inverse and closure theorems below are included with proofs for completeness. The modeling proposal is their coordinated use for production requirements, reusable capability annotations, and development transitions. No priority claim is made for the combination.

There are competing economic interpretations. Starosta [5, pp. 382–385] treats socially necessary software development as participating in the value formation of a relevant mass of subsequent copies. This paper instead conditions the reproduction of an instance on the capabilities already available. Its project account still records development resources. These are different accounting objects, and the algebra does not adjudicate between their interpretations as social value.

The central specification is

$$\boxed{G_C = H_C(I - A_C)^{-1}, \qquad R_C = D_C \odot B_C^*, \qquad \ell_p(C) = h_p + G_C a_p.} \tag{1.1}$$

The first and third equations use ordinary real arithmetic. The second uses Boolean arithmetic. Their meanings, existence conditions, and limits are established below.

2 Production contexts and primitive data

2.1 Notation and units

Write $\mathbb{B} = \{0, 1\}$ with OR, $\vee$, and AND, $\wedge$. For real vectors and matrices, inequalities are entrywise. The notation $w \gg 0$ means that every entry is strictly positive. The vector $\mathbf{e}_j$ is the jth standard basis vector, and I is the identity matrix of the required dimension. For a nonnegative matrix M, $\operatorname{supp}(M) = \mathbf{1}[M > 0]$ is its Boolean support. The spectral radius $\rho(M)$ is the maximum modulus of its complex eigenvalues.

There are $n \geq 1$ selected production nodes, $m \geq 1$ labor categories, and $q \geq 0$ reusable capability types. Write $\mathcal{K} = \{1, \ldots, q\}$ for the capability set, interpreted as empty when $q = 0$, and let $\mathcal{F}$ be a nonempty finite set of producers. Each production node has one specified output unit and one selected recipe. Multiple producers or techniques for the same physical type can be represented by separate nodes when their requirements must remain distinct.

Symbol	Space	Meaning
A_C	$\mathbb{R}_{\geq 0}^{n\times n}$	Ordinary input quantities per unit of selected output.
H_C	$\mathbb{R}_{\geq 0}^{m\times n}$	Direct labor hours by category per unit of output.
G_C	$\mathbb{R}_{\geq 0}^{m\times n}$	Direct and indirect labor requirements.
D_C	$\mathbb{B}^{q\times n}$	Direct capability labels, including current use prerequisites.
B_C	$\mathbb{B}^{n\times n}$	Selected upstream technical dependencies.
R_C	$\mathbb{B}^{q\times n}$	Direct and upstream supporting capability labels.
x, y	$\mathbb{R}_{\geq 0}^{n}$	Gross ordinary output and final demand.
a_p, h_p	$\mathbb{R}_{\geq 0}^{n}, \mathbb{R}_{\geq 0}^{m}$	Whole-project inputs and direct labor.
ω	$\mathbb{R}_{>0}^{m}$	Explicitly supplied common-labor reduction weights.

The entry A_{ij} has units of input i per unit of output j; H_{rj} has units of category-r labor hours per unit of output j. Hence G_{rj} has the same units as H_{rj}, and Gy is a vector of labor hours. Multiplying by $\omega^\top$ is an additional conversion, not an operation already implicit in G.

2.2 Contexts, recipes, and availability

Definition 2.1 (Production context)**.** A context is a record

$$C = (\Omega, t, K, X, \pi). \tag{2.1}$$

The boundary Ω identifies the production system and its suppliers; t identifies the evaluation stage; $K = (K_f)_{f\in\mathcal{F}}$ gives the capability sets accessible to producers; X specifies installed equipment and capacity; and π specifies the selected recipes, suppliers, and output definitions. These data determine the matrices used in (1.1).

A capability is an identified reusable condition of production: for example, a scientific procedure, manufacturing design, usable software release, or runtime. Accessibility need not mean that every producer independently discovered the capability. Sharing and access restrictions belong in K.

For output j, the chosen recipe is (a_j, h_j, d_j), with $a_j \in \mathbb{R}_{\geq 0}^{n}$, $h_j \in \mathbb{R}_{\geq 0}^{m}$, and $d_j \subseteq \mathcal{K}$. The numeric matrices are $A_C = (a_1 \ \cdots \ a_n)$ and $H_C = (h_1 \ \cdots \ h_n)$. Its implementing producer is $f(j)$. Local capability eligibility requires

$$d_j \subseteq K_{f(j)}. \tag{2.2}$$

Equipment availability, material supply, labor availability, and capacity impose additional feasibility conditions. No theorem below eliminates those conditions.

For the static reproduction calculation, the output list is restricted to selected, available recipes and their required suppliers. An item without an available production route has no valuation record under that context. Replacing its absent recipe by a zero column would falsely describe a free production method.

2.3 Standing assumptions

The static model adopts the following assumptions.

A1. Specified outputs and selected routes. Quantities, quality, service duration, suppliers, and recipes are fixed within a context. Alternative routes are evaluated as separate selections; their requirements are not all imposed simultaneously.

A2. Linear ordinary production. Each selected recipe has one output and constant nonnegative input and labor coefficients. Gross output can be divided continuously. Joint production, indivisible batches, congestion, and scale effects require additional structure.

A3. Consistent upstream boundary. All ordinary inputs are represented, or omitted suppliers are accounted for through a consistent boundary reduction as in Appendix A.

A4. Productivity. The selected ordinary production matrix satisfies $\rho(A_C) < 1$. This concerns reproducibility of net output; it is distinct from finite capacity.

A5. Reusable capabilities. A required capability is not itself depleted merely by being used. Any consumable carrier, computing resource, maintenance work, or reproduction activity has its own ordinary inputs and labor.

A6. Conditional development accounting. Historical development is not automatically allocated to every later unit. Current development required by the specified output is represented once, either within its recipe or as a separate project.

These are assumptions of the accounting model. In particular, A6 is not a theorem derived from A1–A5. A finite planning horizon may use an enlarged capability universe or output catalogue; all static results apply to the finite system specified at the evaluation stage.

2.4 Equipment stocks and equipment services

Installed capacity X and per-unit input requirements A are distinct. In a stationary reproduction calculation, a machine coefficient may represent a declared replacement equivalent or an equipment-service input. It need not represent the purchase of a fraction of a machine at each instant. Marx's distinction between continued productive use and gradual value transfer provides relevant background [2, Ch. 8]; the particular coefficient convention remains a modeling choice.

Actual time-ordered equipment purchases, operation, retirement, and inventory use require physical stock accounts. Section 7 therefore distinguishes actual-flow coefficients from stationary replacement allocations.

3 Labor requirements and productive reproduction

The subscript C is suppressed in this section. The context and all coefficients remain fixed.

Theorem 3.1 (Productive reproduction)**.** *For $A \in \mathbb{R}^{n\times n}_{\geq 0}$, the following conditions are equivalent:*

(i) $\rho(A) < 1$*;*
(ii) the matrix series $\sum_{k=0}^{\infty} A^k$ *converges;*
(iii) $I - A$ *is invertible and* $(I - A)^{-1} \geq 0$*;*
(iv) there is a vector $w \gg 0$ *such that* $Aw \ll w$*;*
(v) for every $y \geq 0$*, the equation* $x = Ax + y$ *has a unique nonnegative solution.*

Under these conditions, writing $N = (I - A)^{-1}$ *gives*

$$N = \sum_{k=0}^{\infty} A^k, \qquad x = Ny. \tag{3.1}$$

Proof. To prove (i)⇒(ii), use a complex Jordan representation of A. A Jordan block $J = \lambda I + U$ has nilpotent U, and

$$J^k = \sum_{r=0}^{\min(k,d-1)} \binom{k}{r} \lambda^{k-r} U^r$$

for a block of size d. For $\lambda \neq 0$ and $|\lambda| < 1$, each term is bounded by a polynomial in k times a decaying geometric sequence, so its series converges absolutely. A block with $\lambda = 0$ is nilpotent. There are finitely many blocks, so the matrix series converges. Diagonalizability is not required.

For $S_N = \sum_{k=0}^{N} A^k$, the finite geometric identity gives

$$(I - A)S_N = S_N(I - A) = I - A^{N+1}.$$

Convergence of the series implies $A^{N+1} \to 0$, establishing (ii)⇒(iii) and (3.1). Nonnegativity follows because every term is nonnegative.

For (iii)⇒(iv), set $w = (I - A)^{-1}\mathbf{1}$. Then $w \geq 0$ and $w = Aw + \mathbf{1} \geq \mathbf{1}$, so $w \gg 0$ and $Aw = w - \mathbf{1} \ll w$. For (iv)⇒(i), put $S = \operatorname{diag}(w)$. The row sums of $S^{-1}AS$ are $(Aw)_i/w_i < 1$. Consequently

$$\rho(A) = \rho(S^{-1}AS) \leq \|S^{-1}AS\|_\infty < 1.$$

Condition (iii) implies (v) by inversion. Conversely, apply (v) to $y = \mathbf{1}$; its solution satisfies $x = Ax + \mathbf{1}$, which establishes (iv). □

Theorem 3.2 (Unique occupational labor account)**.** *Let $H \in \mathbb{R}^{m\times n}_{\geq 0}$ and $\rho(A) < 1$. The unique matrix solution of*

$$G = H + GA \tag{3.2}$$

is

$$G = H(I - A)^{-1} = \sum_{k=0}^{\infty} HA^k \geq 0. \tag{3.3}$$

For final demand $y \geq 0$ with feasible reproduction plan $x = Ny$,

$$\ell_C(y) = Gy = Hx. \tag{3.4}$$

For one unit of output j,

$$g_j(C) = G_C\mathbf{e}_j = h_j + G_C a_j. \tag{3.5}$$

Proof. Equation (3.2) is equivalent to $G(I - A) = H$. Theorem 3.1 gives the unique solution (3.3). In the series, H records direct labor, HA the labor in first-level inputs, and HA^k the corresponding kth input layer. Finally, $Gy = HNy = Hx$, and multiplying (3.2) by $\mathbf{e}_j$ gives (3.5). □

The adjective *feasible* in (3.4) matters. For example, a capacity constraint $Mx \leq b$, with specified $M \geq 0$ and $b \geq 0$, must be checked after calculating $x = Ny$. The spectral condition does not imply that an arbitrarily large demand can be supplied at a particular time.

Remark 3.3 (What productivity does and does not say). For $A = [2]$, the matrix $I - A$ is invertible but its inverse is negative, and positive final demand cannot be reproduced. Nevertheless, with $H = [0]$, the labor equation has the nonnegative solution $G = [0]$. Thus $\rho(A) < 1$ is equivalent to the reproduction property in Theorem 3.1, not to the mere existence of a nonnegative G for one arbitrary H. An unused unproductive component may also be removed when a particular demand uses only a productive upstream subsystem.

Proposition 3.4 (Optional scalar reduction)**.** *For supplied weights $\omega \in \mathbb{R}^m_{>0}$, define $v_C = \omega^\top G_C$. It is the unique row-vector solution of*

$$v_C = \omega^\top H_C + v_C A_C, \qquad v_j(C) = \omega^\top G_C \mathbf{e}_j. \tag{3.6}$$

Proof. Multiply (3.2) by $\omega^\top$; uniqueness follows from invertibility of $I - A_C$. □

The units and normalization of ω must be supplied independently. Neither its entries nor a socially valid reduction of skilled to common abstract labor are identified by A and H alone. Retaining G avoids imposing such a reduction prematurely.

3.1 Comparisons, sensitivity, and dimensions

Proposition 3.5 (Monotonicity and coefficient sensitivity)**.** *Suppose* $0 \le A' \le A$, $0 \le H' \le H$, *and* $\rho(A) < 1$, *on the same output and labor coordinates. Then*

$$\rho(A') < 1, \qquad H'(I - A')^{-1} \le H(I - A)^{-1}. \tag{3.7}$$

For differentiable coefficient changes within a fixed feasible regime,

$$dG = (dH + G\,dA)N, \qquad \frac{\partial G_{rj}}{\partial A_{ik}} = G_{ri}N_{kj}. \tag{3.8}$$

If both contexts are productive, the corresponding exact difference is

$$G' - G = \big(H' - H + G(A' - A)\big)(I - A')^{-1}. \tag{3.9}$$

Proof. Nonnegative matrix multiplication preserves entrywise order; induction gives $(A')^k \le A^k$. The nonnegative series for A' converges by domination, and the inverse and labor inequalities follow. Differentiate $G(I - A) = H$ to obtain $dG(I - A) = dH + G\,dA$. For the partial derivative, set dA to the matrix with its only nonzero entry in position (i, k). Subtracting the two fixed-point equations gives $(G' - G)(I - A') = H' - H + G(A' - A)$, proving (3.9). □

Adding an accessible discovery does not imply the inequalities in (3.7). If the chosen A and H remain unchanged, G remains unchanged. New methods may trade one occupation against another, giving incomparable vectors such as $(1, 0)^\top$ and $(0, 1)^\top$. Technique selection needs an objective or a social selection rule outside the accounting identity.

Proposition 3.6 (Invariance under changes of physical units)**.** *Let* S *be positive diagonal, where* S_{ii} *is the number of old units represented by one new unit of output* i. *Define*

$$x' = S^{-1}x, \quad y' = S^{-1}y, \quad A' = S^{-1}AS, \quad H' = HS.$$

Then

$$G' = GS, \qquad G'y' = Gy, \qquad \rho(A') = \rho(A). \tag{3.10}$$

Proof. Similarity preserves eigenvalues, and $(I - A')^{-1} = S^{-1}(I - A)^{-1}S$. Hence $G' = HS\,S^{-1}NS = GS$ and $G'y' = Gy$. Positive unit rescaling also preserves the supports of A and therefore does not change the associated dependency graph. □

4 Reusable capabilities and upstream support

4.1 Current prerequisites and technical dependencies

For compatible Boolean matrices, define

$$(U \odot V)_{ij} = \bigvee_k (U_{ik} \wedge V_{kj}). \tag{4.1}$$

The operator $\odot$ denotes Boolean matrix multiplication throughout this paper, not entrywise multiplication.

Let $(D_C)_{kj} = 1$ exactly when capability k is a direct requirement of recipe j, including prerequisites needed for its current use. If a separate current-use prerequisite graph $U \in \mathbb{B}^{q \times q}$ is desired, with $U_{ab} = 1$ when use of b needs a, then direct labels D^0 are expanded as

$$D = U^* \odot D^0. \tag{4.2}$$

The local set d_j in (2.2) is the support of column j of this expanded matrix. Historical discoveries need not be current-use prerequisites. An already usable binary, for example, need not require its original compiler merely to run. Development dependencies are modeled separately in Section 5.

Use the same n production nodes for A and B. The orientation is

$$B_{ij} = 1 \quad \Longleftrightarrow \quad \text{node } j \text{ depends on upstream node } i \text{ in the selected technical description.} \tag{4.3}$$

Require $\operatorname{supp}(A) \le B$. Extra edges may identify necessary installed-asset or engineering dependencies omitted from per-unit consumption. Their meaning must be declared in the context. Adding entirely new technical nodes requires extending B and D consistently; their dimensions cannot silently diverge.

An extra technical edge is an annotation of the selected support structure. It is not evidence that all knowledge used historically to construct an existing asset is still needed for current operation. Historical lineage may be kept in a separate provenance graph.

Definition 4.1 (Supporting capability annotation)**.** Boolean powers use (4.1), with $B^0 = I$. Define

$$B^* = \bigvee_{k=0}^{n-1} B^k, \qquad R_C = D_C \odot B_C^*, \qquad \mathcal{R}_j(C) = \{k : (R_C)_{kj} = 1\}. \tag{4.4}$$

When $q = 0$, D and R are empty matrices and the capability sets are empty. The ordinary labor equation is unchanged. The same closure definition applies to U with its own dimension, with $U^* = I_0$ in the zero-dimensional case.

Theorem 4.2 (Finite closure and least fixed point)**.** *The matrix $R = D \odot B^*$ is the least solution, in entrywise Boolean order, of*

$$R = D \vee (R \odot B). \tag{4.5}$$

The iteration

$$R^{(0)} = D, \qquad R^{(s+1)} = D \vee (R^{(s)} \odot B) \tag{4.6}$$

stabilizes by $s = n - 1$. No spectral-radius condition on B is required.

Proof. Induction and distributivity show $R^{(s)} = D \odot \bigvee_{k=0}^{s} B^k$. A reachable pair of nodes has a path with no repeated nodes, after cycles are removed, and hence a path of at most $n - 1$ edges. Longer walks add no new reachable pairs. Therefore (4.6) stabilizes at (4.4), which satisfies (4.5). If Z is any solution, then $D \le Z$, and induction in (4.6) gives $R^{(s)} \le Z$ for every s. The limit is the least solution. □

Leastness cannot be replaced by uniqueness. If $D = 0$ and $B = I$, every Boolean R solves (4.5), while the intended least annotation is zero. The closure expresses reachability, so a shared capability is recorded once even when several upstream paths require it.

The set $\mathcal{R}_j(C)$ is not a requirement that producer $f(j)$ personally possess every listed capability. Each producer must satisfy its own local condition (2.2). The final user's machine can be supplied by a producer who possesses the machine-making knowledge.

Proposition 4.3 (When the inverse also encodes support)**.** *If $B = \operatorname{supp}(A)$ and $\rho(A) < 1$, then*

$$\operatorname{supp}((I - A)^{-1}) = B^*, \qquad R = \operatorname{supp}\big(D(I - A)^{-1}\big), \tag{4.7}$$

where the multiplication inside the last support uses real arithmetic. If merely $\operatorname{supp}(A) \le B$, only $\operatorname{supp}(D(I - A)^{-1}) \le D \odot B^*$ is guaranteed.

Proof. Nonnegative multiplication has no cancellation, so $\operatorname{supp}(A^k) = B^k$ in Boolean arithmetic. An entry of the nonnegative Neumann series is positive exactly when at least one corresponding power has a positive entry. This proves the first identity. Applying the same support rule to multiplication by D proves the second. Enlarging the graph can only enlarge reachability. □

For example, let

$$A = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \qquad D = (1 \;\; 0), \qquad B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

The inverse shortcut gives $(1, 0)$, whereas the declared technical dependency gives $R = (1, 1)$. Thus the graph formula in (1.1) is the more general specification.

Definition 4.4 (Contextual valuation record)**.** For a selected available output j, define

$$\boxed{\operatorname{Val}(j \mid C) = (C,\ G_C \mathbf{e}_j,\ \mathcal{R}_j(C)).} \tag{4.8}$$

The term *valuation* names this specified record. Its labor component is a conditional reproduction requirement. Its interpretation as Marxian social value requires the additional economic qualifications in Section 10.

5 Development projects and research prerequisites

5.1 Resource requirements and successful outcomes

A project p has an implementing producer $f(p)$, execution prerequisites d_p, whole-attempt input vector $a_p \in \mathbb{R}^n_{\geq 0}$, direct labor vector $h_p \in \mathbb{R}^m_{\geq 0}$, and realized outcome set $O_p \subseteq \mathcal{K}$. These requirements may depend on the context; $a_p = a_p(C)$ and $h_p = h_p(C)$ are written without arguments when the context is fixed. Execution requires $d_p \subseteq K_{f(p)}$, with current use prerequisites included in d_p. Prerequisites and supply must be available before the result is assumed to exist.

Proposition 5.1 (Conditional project requirement)**.** *If a project is eligible in context C, and its inputs are freshly reproduced under the selected ordinary recipes, its whole-project labor requirement is*

$$\boxed{\ell_p(C) = h_p + G_C a_p.} \tag{5.1}$$

Proof. The ordinary gross output needed for a_p is $x_p = (I - A_C)^{-1} a_p$. Its direct labor is $H_C x_p = G_C a_p$. Add the distinct direct project activity h_p. □

On successful completion, the capability state of the implementing producer changes by

$$K'_{f(p)} = K_{f(p)} \cup O_p. \tag{5.2}$$

Other producers' access changes only through a specified sharing or access transition. An unsuccessful attempt can have $O_p = \varnothing$ while still consuming resources. All actual attempts belong in the chronological account.

Equation (5.2) changes the context. Adoption is a further selection of π' and may then change A, H, B, or D. Discovering an improved process is not identical to installing the needed equipment or adopting it. Projects producing physical by-products or retained prototypes require corresponding physical output and stock entries; (5.1) assumes those effects have been accounted for consistently rather than silently credited twice.

The project requirement is a fresh reproduction measure. If its inputs come from an existing inventory, their historical production is not newly performed at the execution stage. Section 7 gives the corresponding actual-labor account.

5.2 Shared prerequisites

For an explicit dependency calculation, suppose a chosen project catalogue has a correct project-prerequisite graph $P \in \mathbb{B}^{r\times r}$, $r \geq 1$: $P_{ij} = 1$ means executing project j requires the reusable result represented by project i. Each node has a specified result-satisfaction condition, and alternative research routes have already been selected. Every edge includes an access arrangement that makes the predecessor's successful result available to the successor's implementing producer. This simple graph assumes one Boolean satisfaction state per node correctly describes all its selected uses; selective access otherwise requires a more detailed, producer-specific representation.

Let $\tau \in \mathbb{B}^r$ mark target projects, and $b \in \mathbb{B}^r$ mark results that are not already satisfied in the opening context. Traversal stops at a satisfied result; its historical prerequisites need not be recreated.

Proposition 5.2 (Missing-project closure)**.** *Define entrywise AND in the recurrence*

$$u^{(0)} = b \wedge \tau, \qquad u^{(s+1)} = b \wedge \left(\tau \vee (P \odot u^{(s)})\right). \tag{5.3}$$

It stabilizes by $s = r - 1$ at the least fixed point u^*. Its support is exactly the distinct missing projects reachable from a target by following prerequisite links through missing results.

Proof. At round s, the support consists of missing nodes reached by prerequisite paths of length at most s from a missing target, with every traversed node missing. This follows by induction; multiplication by P takes one prerequisite step and the mask b cuts satisfied branches. Repeated-node cycles can be removed, leaving at most $r - 1$ edges. Monotonicity then proves stabilization. Every fixed point contains the initial missing targets and all successive missing prerequisites, establishing leastness. □

If the remaining graph is acyclic, a topological order gives a potential execution order: inductively, every missing predecessor has completed before its successor. This statement is conditional on resource feasibility and successful completion. A dependency cycle among entirely missing results can have a well-defined Boolean closure without an executable first project; it requires an existing seed, a staged bootstrap, or a jointly specified project.

A shared completed prerequisite is represented by one node and is therefore performed once in that plan. This does not erase failed attempts. Nor does the closure solve research-route optimization, alternative substitutes, uncertain outcomes, or the construction of a correct graph for partially available multi-result projects. Such cases require an explicit planner that supplies the dependency description and avoids duplicating one project across its several outputs.

If ordinary matrices A, H and each project's coefficients a_p, h_p remain unchanged throughout the plan, inputs are independently reproduced, and activities are disjoint, the plan requirement is $\sum_{p:u^*_p=1}(h_p + Ga_p)$, where $G = H(I - A)^{-1}$. Eligibility is evaluated in the successive capability contexts. When discoveries alter input-production conditions, evaluate each $\ell_p(C_p)$ in the context C_p applicable to that project. With interstage inventories, use the actual-flow equations below instead of a naive sum of embodied-input charges.

6 Software and ordinary commodities in one model

Specify the output before assigning a recipe. A delivered software copy, a maintained software service, a custom adaptation, and a completed new release are different outputs or activities. Their labor requirements need not coincide.

An existing release s is represented as a reusable capability, so $D_{s,j} = 1$ for a copying or service recipe that requires it. The ordinary recipe includes all current distribution, equipment, energy, hosting, testing,

adaptation, maintenance, or support work required by that specified output. The original code is not represented as one consumed original for every new copy.

Theorem 6.1 (Appending a copying or service activity)**.** *Let (A_0, H_0) be a productive base economy. Add a terminal activity using base inputs $a \geq 0$ and direct labor $h \geq 0$, with its output not used by the base activities:*

$$A = \begin{pmatrix} A_0 & a \\ 0 & 0 \end{pmatrix}, \qquad H = \begin{pmatrix} H_0 & h \end{pmatrix}. \tag{6.1}$$

Then $\rho(A) < 1$ and

$$G = \begin{pmatrix} G_0 & h + G_0 a \end{pmatrix}, \qquad G_0 = H_0(I - A_0)^{-1}. \tag{6.2}$$

Proof. The eigenvalues of the block triangular matrix are those of A_0 together with zero. Direct block multiplication verifies

$$(I - A)^{-1} = \begin{pmatrix} (I - A_0)^{-1} & (I - A_0)^{-1} a \\ 0 & 1 \end{pmatrix}.$$

Multiplication by H gives (6.2). □

Corollary 6.2 (Idealized copying limit)**.** *For a fixed productive base economy, if $a_\varepsilon \to 0$ and $h_\varepsilon \to 0$, then the added activity's unit labor requirement tends to zero:*

$$g_{\mathrm{copy},\varepsilon} = h_\varepsilon + G_0 a_\varepsilon \longrightarrow 0.$$

Its release requirement $D_{s,\mathrm{copy}} = 1$ can remain unchanged.

Proof. The expression is linear and continuous in $(a_\varepsilon, h_\varepsilon)$. □

The limit is a mathematical specialization, not an empirical assertion that actual software delivery or maintained services require no labor. A missing usable release makes the selected copying route unavailable. It does not make its labor requirement zero. The same block theorem applies to a physical terminal product with the specified inputs, so no software-specific value equation is needed.

With fixed coefficients and no additional setup, N_c delivered copies require $N_c g_{\mathrm{copy}}$. If a batch requires setup labor b once, its requirement is $b + N_c g_{\mathrm{copy}}$; applying a per-unit coefficient calibrated at one batch size to a different size would violate A2. Current release development already required by a maintained-service recipe must not be charged again as a separate project.

7 Chronological production and inventories

The static account asks what labor is required to reproduce a specified output in a fixed context. An event-based simulation also needs an account of which activities actually occurred. These questions coincide under suitable conditions, but they are not interchangeable.

Use a common commodity and occupational coordinate system for a finite horizon $t = 0, \ldots, T-1$. Let $s_t \geq 0$ be opening inventories, $x_t \geq 0$ ordinary gross production, $y_t \geq 0$ final deliveries, and $z_t \in \mathbb{N}_0^r$ numbers of project attempts. Let $J_t \in \mathbb{R}_{\geq 0}^{n \times r}$ give the ordinary inputs per attempt and $F_t \in \mathbb{R}_{\geq 0}^{m \times r}$ the direct project labor.

Use actual-flow coefficients $\widehat{A}_t, \widehat{H}_t$ for the ordinary activities. A stationary replacement allocation in A_C must not automatically be interpreted as physical consumption in $\widehat{A}_t$. Physical equipment stocks and their capacity effects must be tracked consistently in X_t.

The inventory and actual-labor accounts are

$$s_{t+1} = s_t + (I - \widehat{A}_t)x_t - y_t - J_t z_t, \tag{7.1}$$

$$L_{\text{actual}} = \sum_{t=0}^{T-1} (\widehat{H}_t x_t + F_t z_t). \tag{7.2}$$

The displayed balances assume project attempts yield capabilities without a separately retained ordinary output. If projects also yield physical outputs, let $W_t \in \mathbb{R}^{n\times r}_{\geq 0}$ record those quantities and add $+W_t z_t$ to (7.1); the reconciliation formulas below then replace J_t by $J_t - W_t$. Actual labor (7.2) is unchanged. This extension accounts for physical flows without prescribing how to allocate a joint project's labor between its results.

All inventories remain nonnegative. Nonstockable service coordinates have inventory fixed at zero. Ineligible activities have zero execution quantities; no valuation is inferred for them from an artificial zero recipe.

The balances must be accompanied by capability eligibility, capacity constraints, and a feasible within-stage schedule. Endpoint balances alone do not establish that a circular production chain can start from the opening stocks. Each activity appears either in ordinary production or among separately accounted project attempts, so labor is not duplicated between $\widehat{H}_t x_t$ and $F_t z_t$.

Project results become available for the next stage unless the current stage is explicitly subdivided:

$$K_{f,t+1} = K_{f,t} \cup \bigcup_{\substack{\text{successful attempts } p \text{ at } t \\ f(p)=f}} O_{p,t}. \tag{7.3}$$

Other access gains and losses need their own specified transitions. Research success probabilities and the rule selecting π_{t+1} are not determined by the accounting equations.

Proposition 7.1 (Reconciliation with reproduction requirements)**.** *Suppose the selected ordinary system for a stage is productive and put $\widehat{G}_t = \widehat{H}_t(I - \widehat{A}_t)^{-1}$. For feasible flows,*

$$\widehat{H}_t x_t + F_t z_t = \widehat{G}_t(y_t + J_t z_t + s_{t+1} - s_t) + F_t z_t. \tag{7.4}$$

If the same feasible catalogue and coefficients $\widehat{A}, \widehat{H}$ apply throughout, then

$$L_{\text{actual}} = \widehat{G}\left(\sum_t y_t + \sum_t J_t z_t + s_T - s_0\right) + \sum_t F_t z_t. \tag{7.5}$$

Proof. Rearrange (7.1) to isolate $(I - \widehat{A}_t)x_t$ and multiply by $\widehat{G}_t$. Since $\widehat{G}_t(I - \widehat{A}_t) = \widehat{H}_t$, this gives (7.4). Under constant $\widehat{G}$, the inventory differences telescope when summed, proving (7.5). □

The vector in parentheses in (7.4) may have negative coordinates when stocks are drawn down; the identity is bookkeeping for feasible flows, not a claim that every such vector is a new nonnegative final demand. With changing contexts, $\sum_t \widehat{G}_t(s_{t+1} - s_t)$ generally cannot be replaced by one terminal inventory adjustment.

Example 7.2 (The smallest inventory double count)**.** One input has $\widehat{A} = 0$ and $\widehat{H} = 1$. A project consumes one input and performs two direct hours. In stage 0, the input is produced and stored, using one hour. In stage 1, the project draws it from stock and uses two hours. Actual labor is $1 + 2 = 3$. Adding stage-0 labor to a fresh project reproduction requirement gives $1 + (2 + 1) = 4$ and counts the input's production twice. Equations (7.1)–(7.2) give three hours.

8 Worked examples

All coefficients in this section are constructed examples, not empirical estimates. A labor unit means one hour of the specified category. Fractions are exact.

8.1 Physical production with researched machinery

Order the outputs as steel, machine, and widget:

$$A = \begin{pmatrix} 0 & 10 & 1 \\ 0 & 0 & 1/100 \\ 0 & 0 & 0 \end{pmatrix}, \qquad H = \begin{pmatrix} 2 & 20 & 1 \end{pmatrix}. \tag{8.1}$$

Here $1/100$ of a machine per widget is a declared stationary reproduction equivalent. Since $A^3 = 0$, $\rho(A) = 0$ and $N = I + A + A^2$. Thus

$$G = \begin{pmatrix} 2 & 40 & 17/5 \end{pmatrix}. \tag{8.2}$$

The machine requires $20 + 10(2) = 40$ hours. The widget requires

$$1 + 1(2) + (1/100)(40) = 17/5 = 3.4 \text{ hours.}$$

Let each recipe have one distinct capability, so $D = I_3$ and $B = \operatorname{supp}(A)$. Then

$$R = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

The widget's support includes steel-making, machine-making, and its own manufacturing capability. Its maker need not personally possess the first two if the inputs are supplied by eligible producers.

8.2 A release, copies, and a software service

Before the release exists, take a base economy consisting of electricity and equipment:

$$A_0 = 0_{2\times 2}, \qquad H_0 = \begin{pmatrix} 1/50 & 100 \end{pmatrix}.$$

A development project requires 1000 units of electricity, $9/5$ units of equipment, and 800 direct hours:

$$\ell_p(C_0) = 800 + 1000(1/50) + (9/5)(100) = 1000. \tag{8.3}$$

This calculation uses the pre-release base context; it does not presume that the development result already exists.

After a successful release and adoption, order outputs as electricity, equipment, delivered copy, and a specified service unit:

$$A = \begin{pmatrix} 0 & 0 & 1/100 & 10 \\ 0 & 0 & 1/100000 & 1/1000 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad H = \begin{pmatrix} 1/50 & 100 & 11/1250 & 1/5 \end{pmatrix}. \tag{8.4}$$

As $A^2 = 0$, direct calculation yields

$$G = \begin{pmatrix} 1/50 & 100 & 1/100 & 1/2 \end{pmatrix}. \tag{8.5}$$

In particular,

$$\begin{aligned} g_{\text{copy}} &= 11/1250 + (1/100)(1/50) + (1/100000)(100) = 1/100, \\ g_{\text{service}} &= 1/5 + 10(1/50) + (1/1000)(100) = 1/2. \end{aligned}$$

The release's direct capability row is $D_s = (0, 0, 1, 1)$. Its presence does not require a consumed-code input. With $B = \text{supp}(A)$ this row also remains $(0, 0, 1, 1)$ in R.

With fresh, disjoint inputs and no additional setup, this development followed by 1000 copies requires $1000 + 1000(1/100) = 1010$ hours. The copies' conditional unit requirement after development is still $1/100$ hour. Both numbers answer well-defined questions; a universal historical allocation rule is not needed to obtain them.

8.3 Two occupations without an assumed wage conversion

For raw material and a finished product, use production labor and engineering labor:

$$A = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}, \qquad H = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}, \qquad G = \begin{pmatrix} 3 & 7 \\ 0 & 2 \end{pmatrix}.$$

A finished product requires seven production hours and two engineering hours. If $\omega = (1, 3)^\top$ is independently supplied, then $\omega^\top G = (3, 13)$. The value 13 follows from the chosen reduction weights; it is not a wage ratio derived by the input-output calculation.

8.4 Why materials accumulate while knowledge is shared

Consider a diamond-shaped chain: node 1 supplies both nodes 2 and 3, which both supply node 4.

$$A = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad H = (1, 0, 0, 0).$$

Then $G = (1, 1, 1, 2)$. Final output 4 consumes two distinct units of input 1, so both units' labor is required. For one capability attached to node 1, let $D = (1, 0, 0, 0)$ and $B = \text{supp}(A)$. Boolean closure gives $R = (1, 1, 1, 1)$: output 4 requires that capability once as a condition, even though two paths depend on it. Removing the duplicated material quantity would be an error; duplicating the reusable capability label would add no information.

8.5 A shared research prerequisite

Let p_0 create α with four direct hours, p_1 require α and create β with six, and p_2 require α and create γ with eight. Take no ordinary project inputs and no change in these labor requirements. For targets β, γ and no initial results,

$$P = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad \tau = (0, 1, 1)^\top, \qquad b = (1, 1, 1)^\top.$$

The missing-project closure is $u^* = (1,1,1)^\top$, so the labor requirement is $4+6+8=18$. Adding two complete chains would give 22 and repeat p_0. If α is initially accessible, $b=(0,1,1)^\top$ and the requirement is 14; its development history need not be recreated.

9 Computation and reproducibility

9.1 Numeric requirements

For numerical implementation, solve

$$(I - A_C)^\top G_C^\top = H_C^\top \tag{9.1}$$

instead of explicitly forming an inverse. A dense factorization has arithmetic cost $O(n^3)$, followed by $O(mn^2)$ for the m right-hand sides. Sparsity may reduce actual work, but the reduction depends on the graph and factorization.

For a single final demand, it is sufficient to solve $(I - A_C)x = y$ and compute $H_C x$. If only a scalar reduction is needed, solve one transposed system with right-hand side $H_C^\top \omega$. None of these operations determines whether the selected route is economically or physically admissible; those checks precede valuation.

Proposition 9.1 (A certified truncation bound)**.** *Suppose $w \gg 0$ satisfies $Aw \ll w$. Let $S = \operatorname{diag}(w)$ and $\alpha = \|S^{-1}AS\|_\infty < 1$. For $G^{[k]} = H\sum_{i=0}^{k} A^i$,*

$$\left\|(G - G^{[k]})S\right\|_\infty \le \|HS\|_\infty \frac{\alpha^{k+1}}{1-\alpha}. \tag{9.2}$$

Proof. Write $\widetilde{A} = S^{-1}AS$. Then $(G - G^{[k]})S = HS\sum_{i=k+1}^{\infty} \widetilde{A}^i$. Submultiplicativity and the geometric-series bound give (9.2). □

A large unscaled row sum is not evidence of nonproductivity: the steel-machine example has a row sum of 11 but is nilpotent. In floating-point calculations, thresholding an inverse to recover graph reachability can also lose small positive paths. Use the explicitly stored technical graph for R.

9.2 Capability propagation

A simple exact Boolean algorithm is

initialize	$R \leftarrow D$;
repeat at most $n-1$ times	$R \leftarrow D \vee (R \odot B)$;
return	R.

Theorem 4.2 proves correctness. Early termination is valid when an iteration makes no change.

For sparse graphs, identify strongly connected components, take the union of direct labels within each component, and propagate these sets through the resulting acyclic graph in upstream-to-downstream order. Every node in a component reaches every other node there, so its accumulated support is the same; the acyclic propagation then adds precisely the reachable predecessor labels. With e edges, capability bitsets, and machine word size w_{bit}, this requires $O((n+e)(1+\lceil q/w_{\text{bit}}\rceil))$ word operations under the standard constant-time word model, including graph traversal and label propagation.

Research-project closure uses (5.3) with the availability mask. Detecting an unsatisfied prerequisite cycle is a planning check; it is not repaired by computing more closure iterations.

9.3 Reproducibility protocol

All worked examples specify their full matrices or project coefficients and are reproducible with rational arithmetic. For an implementation, verify the identities $G(I - A) = H$ and $R = D \vee (R \odot B)$, together with leastness through initialization at D. The latter condition matters because a fixed-point residual alone does not establish the intended Boolean solution.

A reproducible evaluation record should retain: the context identifier; output units and quality; selected suppliers and recipes; matrix data; local capability eligibility; the definition of any extra technical edges; the labor categories; any scalar weights; and the treatment of equipment services. For chronological results, retain inventories, project attempts, realized outcomes, and activity schedules. No empirical data are needed to reproduce the constructed examples.

10 Interpretation, limits, and conclusions

The model establishes consistency of three related accounts: conditional ordinary reproduction, logical capability support, and resource-consuming development. It does not establish that every historical expenditure becomes socially necessary labor, nor that a firm's own technical requirement is automatically the prevailing social value. A factory, national, or global context must specify the suppliers and production conditions being represented. Appendix A shows how exact upstream elimination preserves the full model's labor requirements when a smaller reporting boundary is used.

Interpreting $v_j(C)$ as a scalar Marxian social value requires an appropriate social production context and a justified reduction to common abstract labor. Market validation, technique prevalence, quality comparability, and the definition of the relevant product remain economic questions. The equations alone do not determine prices, wages, profits, rents, exploitation ratios, or the redistribution of surplus value through exchange.

The choice in A6 should be assessed as a choice of accounting object. The project account makes development labor visible, while the instance account conditions reproduction on existing capability. Allocating some project burden across a specified cohort of products is possible as a separate reporting rule, but its allocation basis and horizon must be supplied. It is not implied by the inverse or by Boolean closure. Likewise, a low copying requirement does not by itself identify all software revenue as rent or decide the social status of programmers' labor.

The ordinary production model uses fixed coefficients, continuous quantities, and a selected route. It does not solve joint production, scale economies, batch indivisibilities, stochastic discovery, endogenous technical choice, or the detailed dynamics of equipment aging. The capability account assumes correct labels and dependencies; an incorrect graph can yield a mathematically correct closure of incorrect data. The numerical examples illustrate internal consistency and are not empirical validation.

Subject to those stated assumptions, physical production and software share one quantitative equation. The separate Boolean annotation preserves the reusable context of that equation, including the research supporting upstream machinery. Research itself remains an activity with inputs, labor, prerequisites, and outcomes. The concise formal record is therefore

$$\boxed{\begin{array}{cc} G_C = H_C(I - A_C)^{-1}, & R_C = D_C \odot B_C^*, \\ \ell_p(C) = h_p + G_C a_p, & \mathrm{Val}(j \mid C) = (C, G_C \mathbf{e}_j, \mathcal{R}_j(C)). \end{array}}$$

A Exact changes of the reporting boundary

A smaller reporting boundary must preserve the upstream requirements it omits. Partition the full productive system into nonempty sets of retained nodes L and external nodes E:

$$A = \begin{pmatrix} A_{LL} & A_{LE} \\ A_{EL} & A_{EE} \end{pmatrix}, \qquad H = \begin{pmatrix} H_L & H_E \end{pmatrix}. \tag{A.1}$$

Thus A_{EL} records external inputs used by retained production, and A_{LE} records retained inputs used by external production. Both directions can be present.

Proposition A.1 (Exact numeric boundary elimination)**.** *If $\rho(A) < 1$, let $N_E = (I - A_{EE})^{-1}$ and define*

$$A_{\text{eff}} = A_{LL} + A_{LE} N_E A_{EL}, \qquad H_{\text{eff}} = H_L + H_E N_E A_{EL}. \tag{A.2}$$

Then $\rho(A_{\text{eff}}) < 1$, and the retained columns of the full labor matrix satisfy

$$G_L = H_{\text{eff}} (I - A_{\text{eff}})^{-1}. \tag{A.3}$$

Proof. Choose $w = (w_L, w_E) \gg 0$ with $Aw \ll w$, using Theorem 3.1. The external block implies $A_{EE} w_E < w_E$, so N_E exists and is nonnegative. It also gives $(I - A_{EE}) w_E > A_{EL} w_L$. Since $N_E \geq I$, multiplication preserves strict positivity and yields $w_E > N_E A_{EL} w_L$. The retained block consequently gives

$$w_L > A_{LL} w_L + A_{LE} w_E \geq (A_{LL} + A_{LE} N_E A_{EL}) w_L.$$

The same theorem implies $\rho(A_{\text{eff}}) < 1$.

The block labor equations give $G_E = (H_E + G_L A_{LE}) N_E$ and $G_L = H_L + G_L A_{LL} + G_E A_{EL}$. Substitution yields $G_L = H_{\text{eff}} + G_L A_{\text{eff}}$, proving (A.3). □

The term H_{eff} is an effective boundary labor coefficient, not exclusively direct labor performed inside L. When external production uses no retained inputs, $A_{LE} = 0$ and the rule reduces to adding the external labor content of imports. Setting that content to zero would change the accounting object.

Formula (A.3) gives the retained columns of the full labor account. For a general final demand (y_L, y_E), external final demand must also be included:

$$x_L = (I - A_{\text{eff}})^{-1} (y_L + A_{LE} N_E y_E), \qquad \ell = H_{\text{eff}} x_L + H_E N_E y_E. \tag{A.4}$$

These identities follow by eliminating the external block of $x = Ax + y$ and then substituting into Hx. Removing external equations does not remove external capacity or access constraints; feasibility still refers to the full system.

For capability annotations, partition B and $D = (D_L \; D_E)$ in the same way and define

$$B_{\text{eff}} = B_{LL} \vee (B_{LE} \odot B_{EE}^* \odot B_{EL}), \tag{A.5}$$

$$D_{\text{eff}} = D_L \vee (D_E \odot B_{EE}^* \odot B_{EL}). \tag{A.6}$$

Proposition A.2 (Exact Boolean boundary elimination)**.** *The retained columns of the full capability annotation satisfy*

$$R_L = D_{\text{eff}} \odot B_{\text{eff}}^*. \tag{A.7}$$

Proof. A path to a retained node can begin at a retained labeled node, or at an external labeled node and first enter L through an external-only path followed by an E-to-L edge. These cases give D_{eff}. Between subsequent visits to retained nodes, a path either uses a retained-to-retained edge or makes an excursion through external nodes, giving B_{eff}. Decomposing every full path this way, and expanding every effective path back into full paths, proves equality of the reachable labels. □

Effective labels summarize upstream support. They must not become a local eligibility test requiring retained producers personally to possess every external supplier's capability. Exact boundary elimination preserves one specified full context; it does not equate a firm's methods with socially prevailing methods. The effective matrices must be recomputed when the omitted production conditions change.

B Scope of the mathematical results

The results require only finite-dimensional real linear algebra, nonnegative coefficients, and finite graph reachability. They do not require irreducibility or diagonalizability. The selected-route model does not require a theory of equilibrium prices, but it also does not select among alternative production or research routes. The spectral hypothesis is needed for the stated universal reproduction account; Boolean closure is finite even on cyclic graphs.

For clarity, the manuscript contains three different kinds of statements:

1. **Definitions and conventions:** the context record, selected recipes, reusable capability representation, and conditional historical-development rule.
2. **Mathematical consequences:** existence and uniqueness of the labor account, least capability closure, sensitivity and unit invariance, software specialization, project closure, inventory reconciliation, and boundary elimination.
3. **External economic choices:** the empirical coefficients, socially relevant context, common-labor reduction, research outcomes, technology adoption, market validation, and any historical cost-allocation rule.

A proof in the second category does not validate a choice in the first or identify an unknown in the third. The examples are fully specified synthetic calculations. This working paper supplies a formal specification and proofs, without claiming empirical calibration, peer review, or priority over all related models.

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C Conditional labor-type conversion

Revision scope. This appendix supplies an optional closure for the weights ω in Proposition 3.4 and Example 8.3. Set $\omega = \alpha_C$ when adopting the closure below. The original occupational requirements G_C, capability annotations R_C, project accounts, and inventory identities remain unchanged. A new economic assumption determines the conversion; it is not implied by the earlier equations. Sections C.9–C.12 complete the reporting-boundary and time-unit rules, make the identification limit explicit, and document the audit.

C.1 Additional primitives and units

Fix a productive context C with n outputs and m labor types. All labor types are measured in *actual clock hours*; differences between types are not already built into H_C. Retain

$$G_C = H_C(I - A_C)^{-1}, \qquad A_C \geq 0, \quad H_C \geq 0, \quad \rho(A_C) < 1. \tag{C.1}$$

Introduce the following nonnegative data, on the same output and occupational coordinates.

Symbol	Dimension	Meaning and units
S_C	$n \times m$	Subsistence and recurrent reproduction goods or services per productive hour of each labor type. Entry S_{ij} is units of output i per type-j hour.
E_C	$n \times m$	One-time education and training goods or services per newly qualified worker of type j.
U_C	$m \times m$	Additional direct training labor per newly qualified worker: U_{ij} is type-i hours per type-j worker, excluding labor already in $G_C E_C$.
Γ_C	$m \times m$	$\mathrm{diag}(\gamma_1, \ldots, \gamma_m)$, with $0 < \gamma_j < \infty$ productive lifetime hours per newly qualified worker.
b	$\{1, \ldots, m\}$	Reference labor type used to normalize $\alpha_b = 1$.

Here U_C denotes *numeric training labor*; it is not the Boolean current-use prerequisite graph U of Section 4.1. Likewise M_C, introduced below, is a labor-reproduction matrix, not the example capacity-constraint matrix in Section 3. All symbols in this appendix have the meanings explicitly given here.

More fully, the extended context is

$$\widehat{C} = (C, S_C, E_C, U_C, \Gamma_C, b).$$

We suppress the hat for brevity. These additional data are specified independently of the unknown conversion vector. "Endogenous" below means determined *conditional on these data and the closure assumption*; it does not mean inferred from A_C, H_C alone.

The term *reproduction* here refers to a declared social provisioning and qualification account, not a biological characterization or a valuation of persons. The categories, consumption baskets, training arrangements, and lifetime horizon are modeling choices.

C.2 The reproduction matrix and the additional axiom

For one productive hour of type j, the subsistence bundle requires $G_C S_C[:, j]$ occupational hours. Education and additional training require $G_C E_C[:, j] + U_C[:, j]$ per newly qualified worker. Under the declared lifetime convention, divide the latter vector by γ_j. Thus define

$$\boxed{M_C = G_C S_C + (G_C E_C + U_C)\Gamma_C^{-1}.} \tag{C.2}$$

Entry M_{ij} is type-i reproduction labor per productive type-j hour. Equivalently, with

$$F_C = S_C + E_C\Gamma_C^{-1}, \qquad T_C = U_C\Gamma_C^{-1}, \qquad M_C = G_C F_C + T_C, \tag{C.3}$$

F_C describes reproduction goods per hour and T_C the remaining direct training labor per hour.

Accounting conventions. The two terms in M_C must represent disjoint activities. Training-period subsistence belongs in E_C if it is not already included in the recurrent account. When a purchased education service in E_C already includes its teachers' labor through G_C, that same teaching must not also appear in U_C. Unpaid work can be represented when explicitly included; omitted work is not automatically zero in reality. Foregone earnings are not labor inputs. Trainees' hours require an explicit productive-activity convention and must not also be counted as hours delivered to another activity.

Lifetime allocation is a stationary cohort or specified planning convention, not a theorem about worker histories. With heterogeneous careers, entries must be calibrated consistently as cohort totals per total productive hours; an arbitrary average of individual ratios need not give the same result. Continuing training is included once in the chosen recurrent or project account.

Let $\alpha_C \gg 0$ convert one clock hour of each type into common-labor units. The modeled reproduction burden of a type-j hour is

$$c_j^{\text{rep}} = \alpha_C^{\mathsf{T}} M_C[:, j] = (M_C^{\mathsf{T}}\alpha_C)_j.$$

This burden does not by itself identify the conversion weight of that hour.

Assumption C.1 (Equal residual contribution)**.** There is a scalar $\kappa_C > 0$ such that

$$\boxed{\alpha_C - M_C^{\mathsf{T}}\alpha_C = \kappa_C \mathbf{1}, \qquad \alpha_{C,b} = 1.} \tag{C.4}$$

The residual after the specified reproduction burden is equal for one clock hour of every labor type.

This is the substantive closure. It neither follows from $G = H + GA$ nor identifies an empirical marginal contribution, a wage, or an exploitation rate. In particular, α_j and c_j^{rep} are *different* objects. The normalization fixes the common unit; the axiom fixes the relative weights. Actual employment, labor availability, cohort replacement, and capacity still require separate feasible schedules.

C.3 Existence, uniqueness, and ratios

Theorem C.2 (Normalized positive conversion)**.** *Let $M \geq 0$ be $m \times m$ and let b be a reference type. A pair (α, κ) with $\alpha \gg 0$, $\kappa > 0$, and $\alpha_b = 1$ satisfies (C.4) if and only if $\rho(M) < 1$. In that case the pair is unique and*

$$\beta = (I - M^{\mathsf{T}})^{-1}\mathbf{1}, \qquad \boxed{\alpha = \frac{\beta}{\beta_b}, \qquad \kappa = \frac{1}{\beta_b}.} \tag{C.5}$$

No irreducibility or diagonalizability hypothesis is needed.

Proof. If $\rho(M) < 1$, the nonnegative inverse exists by Theorem 3.1 applied to M^{T}. Its Neumann series gives $\beta = \sum_{r \geq 0} (M^{\mathsf{T}})^r \mathbf{1} \geq \mathbf{1} \gg 0$. The displayed formulas satisfy every requirement. Conversely, a positive solution gives $M^{\mathsf{T}}\alpha = \alpha - \kappa\mathbf{1} \ll \alpha$, so the positive-vector criterion in Theorem 3.1 implies $\rho(M^{\mathsf{T}}) = \rho(M) < 1$. Finally, every solution obeys $\alpha = \kappa\beta$, and normalization forces $\kappa\beta_b = 1$. □

The hypothesis on M is *additional* to $\rho(A) < 1$. For example, $A = [0]$, $H = [1]$, $S = [1]$, and $E = U = 0$ give a productive goods system but $M = [1]$, for which no strictly positive residual is possible. With $M = [2]$, inversion alone yields a negative β, which is inadmissible. Spectral admissibility is a property of this closure, not a proof that the actual economy is physically or socially viable.

Definition C.3 (Pairwise labor-type ratio)**.** For types i, j, define

$$\boxed{r_{ij}(C) = \frac{\alpha_i(C)}{\alpha_j(C)} = \frac{\beta_i(C)}{\beta_j(C)}.} \tag{C.6}$$

Thus one type-i hour receives r_{ij} times the common-labor weight of one type-j hour.

Proposition C.4 (Normalization and special cases)**.** *Changing the reference from b to c gives $\alpha^{(c)} = \alpha^{(b)}/\alpha_c^{(b)}$ and leaves every r_{ij} unchanged. If $M = 0$, all weights are one. More generally, if every column sum of M equals $s < 1$, all normalized weights are one. With one labor type, its normalized weight is always one whenever the closure is admissible.*

Proof. Both reference choices divide the same β by one of its entries. If $M^{\mathsf{T}}\mathbf{1} = s\mathbf{1}$, then $\beta = (1 - s)^{-1}\mathbf{1}$, which proves the remaining statements. □

Identification limit. Replacing the common residual by κq, for any specified $q \gg 0$, gives $\alpha(q) \propto (I - M^{\mathsf{T}})^{-1} q$. The same reproduction data can therefore yield different ratios under different residual assumptions. Equal residuals mean specifically $q = \mathbf{1}$; the resulting ratios are not uniquely forced by reproduction data without that axiom.

C.4 Scalar commodity values and a unified block system

Use a *column* vector for scalar output values:

$$\boxed{v_C = G_C^{\mathsf{T}}\alpha_C, \qquad v_j(C) = \alpha_C^{\mathsf{T}} G_C e_j.} \tag{C.7}$$

This is the transpose of the row-vector convention in Proposition 3.4, with $\omega = \alpha_C$. For fixed α_C, it is the unique solution of

$$v_C = H_C^{\mathsf{T}}\alpha_C + A_C^{\mathsf{T}} v_C.$$

The labor-type ratio r_{ij} is not a ratio of commodity values; i, j in (C.6) index labor types.

Proposition C.5 (Equivalent augmented system)**.** *Suppress the context subscript and use F, T from (C.3). Form the nonnegative requirement matrix*

$$\mathcal{A} = \begin{pmatrix} A & F \\ H & T \end{pmatrix}. \tag{C.8}$$

Because the standing assumption is $\rho(A) < 1$, the matrix $M = H(I - A)^{-1}F + T$ is well-defined. Then $\rho(\mathcal{A}) < 1$ if and only if $\rho(M) < 1$. Under these conditions the unified equations are

$$\begin{pmatrix} v \\ \alpha \end{pmatrix} = \mathcal{A}^{\mathsf{T}} \begin{pmatrix} v \\ \alpha \end{pmatrix} + \begin{pmatrix} 0 \\ \kappa \mathbf{1} \end{pmatrix}, \qquad \alpha_b = 1. \tag{C.9}$$

Eliminating the commodity block yields exactly (C.4) and (C.7).

Proof. If $\rho(\mathcal{A}) < 1$, choose $(u, z) \gg 0$ with $\mathcal{A}\binom{u}{z} \ll \binom{u}{z}$. Then $Au \ll u$, so $N = (I - A)^{-1} \geq 0$. The first block gives $u - NFz = N(u - Au - Fz) \gg 0$, since $N \geq I$; the second gives $Hu + Tz < z$. Consequently $(HNF + T)z < z$, proving $\rho(M) < 1$.

Conversely, if both spectral conditions hold, set $N = (I - A)^{-1}$ and $Z = (I - M)^{-1}$. Direct multiplication verifies

$$(I - \mathcal{A})^{-1} = \begin{pmatrix} N + NFZHN & NFZ \\ ZHN & Z \end{pmatrix} \geq 0. \tag{C.10}$$

Theorem 3.1 then gives $\rho(\mathcal{A}) < 1$. In (C.9), the first block is $v = A^{\mathsf{T}}v + H^{\mathsf{T}}\alpha$, hence $v = G^{\mathsf{T}}\alpha$. Substituting in the second block gives $\alpha = (GF + T)^{\mathsf{T}}\alpha + \kappa\mathbf{1}$, as claimed. □

This block representation is a formal requirement account. It does not assert that labor hours are physical commodities or that the augmented inverse measures actual clock hours. There is no need to construct the larger matrix computationally: solve the ordinary system for G, then the m-dimensional system for β.

C.5 Context changes, invariance, and sensitivity

Proposition C.6 (Physical-unit invariance)**.** *Let P be a positive diagonal change of commodity units as in Proposition 3.6. Set*

$$A' = P^{-1}AP, \quad H' = HP, \quad S' = P^{-1}S, \quad E' = P^{-1}E,$$

and keep U, Γ, b fixed. Then $G' = GP$, $M' = M$, $\alpha' = \alpha$, and $v' = Pv$. For $y' = P^{-1}y$, $v'^{\mathsf{T}}y' = v^{\mathsf{T}}y$.

Proof. Use Proposition 3.6 and substitute into (C.2). All inserted P factors cancel, so $M' = M$. The normalized solution is unique. Finally $v' = (GP)^{\mathsf{T}}\alpha = Pv$ and the scalar total is unchanged. □

Labor-unit convention. Equal residuals are per clock hour. Renaming one occupation's unit from an hour to a day requires changing its residual component as well as its coefficients. Keeping $\mathbf{1}$ unchanged after an unequal labor-unit rescaling would impose a different axiom, not merely a new notation. Section C.10 proves the precise transformation.

Proposition C.7 (Sensitivity within a fixed admissible regime)**.** *Let $N = (I - A)^{-1}$ and $W = (I - M^{\mathsf{T}})^{-1}$. For differentiable coefficient changes on unchanged coordinates,*

$$\begin{aligned} \mathrm{d}G &= (\mathrm{d}H + G\,\mathrm{d}A)N, \\ \mathrm{d}M &= (\mathrm{d}G)F + G\,\mathrm{d}F + \mathrm{d}T, \\ \mathrm{d}\beta &= W(\mathrm{d}M)^{\mathsf{T}}\beta, \\ \mathrm{d}\alpha &= \frac{(I - \alpha e_b^{\mathsf{T}})\,\mathrm{d}\beta}{\beta_b}, \\ \mathrm{d}v &= (\mathrm{d}G)^{\mathsf{T}}\alpha + G^{\mathsf{T}}\mathrm{d}\alpha. \end{aligned} \tag{C.11}$$

Unnormalized β is entrywise nondecreasing in M on productive nonnegative systems. Normalized weights and their ratios need not be entrywise nondecreasing.

Proof. The first identity is Proposition 3.5. Differentiate $M = GF + T$, $(I - M^{\mathsf{T}})\beta = \mathbf{1}$, $\alpha = \beta/\beta_b$, and $v = G^{\mathsf{T}}\alpha$. For monotonicity, compare the nonnegative Neumann series of two ordered M matrices. Division by the changing positive entry β_b does not preserve such an ordering in general. □

Research can change ordinary recipes, training methods, subsistence requirements, or productive lifetimes. Consequently both G_C and α_C can change. A discovery unused in every selected ordinary and reproduction activity leaves these quantities unchanged. A decrease in direct engineering hours need not imply a proportionate decrease in a normalized scalar value, because the weights may also change.

Comparisons over time must retain the same labor categories, reference convention, and stated output quality, or supply explicit mappings. A historical weighted total using α_{C_t} at each stage is not automatically equal to revaluing all past labor with one final-period vector. Keep the occupational ledger when later revaluation may be needed.

C.6 A fully specified exact example

All numbers below are synthetic. Take two labor types, production and engineering, and six outputs: a consumption bundle, learning materials, raw material, a physical product, a delivered software copy, and a hosted-service unit. To isolate the conversion, use the simplified selected recipes

$$A_{34} = 2, \quad A_{ij} = 0 \text{ otherwise}, \qquad H = \begin{pmatrix} 1 & 0 & 3 & 1 & 1/100 & 1/5 \\ 0 & 1 & 0 & 2 & 1/500 & 1/10 \end{pmatrix}. \tag{C.12}$$

These are constructed linear recipes, not estimated sector accounts. In particular the simplified copying and service columns omit other inputs. They may be replaced by detailed supplier chains without changing the formal procedure. Since $A^2 = 0$,

$$G = H(I + A) = \begin{pmatrix} 1 & 0 & 3 & 7 & 1/100 & 1/5 \\ 0 & 1 & 0 & 2 & 1/500 & 1/10 \end{pmatrix}. \tag{C.13}$$

The physical-product subnetwork reproduces the seven production hours and two engineering hours of Example 8.3.

Specify recurrent and qualification inputs as

$$S = \begin{pmatrix} 1/5 & 1/5 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}, \qquad E = \begin{pmatrix} 0 & 2000 \\ 400 & 3000 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}, \qquad U = \begin{pmatrix} 0 & 0 \\ 600 & 2000 \end{pmatrix}, \qquad \Gamma = 20000 I_2. \tag{C.14}$$

Both types receive one-fifth of a consumption bundle per productive hour. The learning-material inputs in E and additional instructor time in U are distinct activities. The declared productive lifetime is 20,000 hours for each type. Training-period consumption for engineering is included in E and not repeated in S.

The two reproduction components are

$$GS = \begin{pmatrix} 1/5 & 1/5 \\ 0 & 0 \end{pmatrix}, \qquad (GE + U)\Gamma^{-1} = \begin{pmatrix} 0 & 1/10 \\ 1/20 & 1/4 \end{pmatrix}.$$

Hence

$$\boxed{M = \begin{pmatrix} 1/5 & 3/10 \\ 1/20 & 1/4 \end{pmatrix}.} \tag{C.15}$$

Its eigenvalues are $1/10$ and $7/20$. Therefore $\rho(M) = 7/20 < 1$, as required. Neither the goods-production identity nor the reproduction-matrix calculation uses wage rates.

For software, the existing release is still required in D as in Section 6; the zero material-input coefficients in this constructed example do not imply that an unavailable release can be copied. All selected routes and training activities must satisfy the original capability and capacity conditions.

C.7 Example results: goods, software, services, and projects

Using production labor as the reference type, exact inversion gives

$$\beta = \begin{pmatrix} 160/117 \\ 220/117 \end{pmatrix}, \qquad \boxed{\alpha = \begin{pmatrix} 1 \\ 11/8 \end{pmatrix}, \quad \kappa = 117/160, \quad r_{\text{eng,prod}} = 11/8.} \tag{C.16}$$

The reproduction burden and hour-weight differ:

$$M^{\mathsf{T}}\alpha = \begin{pmatrix} 43/160 \\ 103/160 \end{pmatrix}, \qquad \alpha - M^{\mathsf{T}}\alpha = \frac{117}{160}\begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Thus one engineering hour receives 1.375 reference-hour-equivalents *under this closure and these coefficients*; its modeled reproduction burden is 103/160, not 11/8. No empirical claim about engineering follows from this synthetic example.

Output	Occupational requirement $(g_{\text{prod}}, g_{\text{eng}})$	Scalar value	Decimal
Consumption bundle	$(1, 0)$	1	1
Learning materials	$(0, 1)$	11/8	1.375
Raw material	$(3, 0)$	3	3
Physical product	$(7, 2)$	39/4	9.75
Delivered copy	$(1/100, 1/500)$	51/4000	0.01275
Hosted-service unit	$(1/5, 1/10)$	27/80	0.3375

Every scalar entry is the same operation $v_j = \alpha^{\mathsf{T}} g_j$. The earlier illustrative value 13 in Example 8.3 used the externally supplied weights $(1, 3)^{\mathsf{T}}$. The value 39/4 here instead follows from the present reproduction data and equal-residual assumption. The physical labor vector $(7, 2)^{\mathsf{T}}$ has not changed.

Research projects. For an eligible project with fresh, disjoint inputs, the scalar counterpart of Proposition 5.1 is

$$\boxed{V_p(C) = \alpha_C^{\mathsf{T}} \ell_p(C) = \alpha_C^{\mathsf{T}} h_p + v_C^{\mathsf{T}} a_p.} \tag{C.17}$$

For $h_p = (10, 20)^{\mathsf{T}}$ and $a_p = (5, 2, 0, 0, 0, 0)^{\mathsf{T}}$, one gets $\ell_p = (15, 22)^{\mathsf{T}}$ and $V_p = 181/4 = 45.25$. This is a weighted project account; it is not an additional automatic charge on every later copy. With existing inventory, retain the actual-flow distinction in Section 7.

No duplicate labor ledger. Gy is an occupational labor vector; $\alpha^{\mathsf{T}} Gy$ is a chosen scalarization. Do not append M-generated reproduction rounds to the actual-labor ledger and then also apply these weights as if both were new work. Actual subsistence and training activities, when produced, are counted once through their explicit recipes or project activities. Development of a reusable teaching method belongs to the context/project account; instruction required anew for each cohort belongs to E or U. This distinction also applies to software-based training.

C.8 Computation, reporting, and interpretive limits

Implementation. First compute G by solving $(I - A)^{\mathsf{T}} G^{\mathsf{T}} = H^{\mathsf{T}}$. Form M from (C.2), verify its nonnegative productive regime, and solve

$$(I - M^{\mathsf{T}})\beta = \mathbf{1}, \qquad \alpha = \beta/\beta_b, \qquad v = G^{\mathsf{T}}\alpha. \tag{C.18}$$

Do not form either inverse merely to obtain these solutions. A strictly positive candidate β with a certified strictly positive vector $(I - M^{\mathsf{T}})\beta$ supplies a productivity certificate by Theorem 3.1. Floating-point residuals should be scaled and checked with error bounds; a small unscaled residual near singularity is not a certificate of a reliable ratio.

Check $G(I - A) = H$, $(I - M^{\mathsf{T}})\beta = \mathbf{1}$, $\alpha_b = 1$, $(I - M^{\mathsf{T}})\alpha = \kappa\mathbf{1}$, and $(I - A^{\mathsf{T}})v = H^{\mathsf{T}}\alpha$. The preceding example was checked with exact rational arithmetic, including its block inverse, reference change, physical-unit rescaling, and project scalarization. Nonproductive one-type counterexamples and a different residual vector were also checked.

Reporting. Retain (S, E, U, Γ, b) and the closure assumption with the original context. Store the occupational vector as well as its scalarization:

$$\mathrm{Val}_\alpha(j \mid \widehat{C}) = (\widehat{C},\ G_C e_j,\ \alpha_C,\ v_j(C),\ \mathcal{R}_j(C)).$$

$\mathcal{R}_j$ continues to describe the output's production chain. The data and capabilities supporting training and the conversion weights have separate provenance; they must not become local runtime requirements of a software user or final producer. The original Boolean equation is unchanged.

Limits. The formula does not equate wage ratios with labor-type ratios. If subsistence inputs are estimated from wage-financed consumption, that empirical dependence must be acknowledged even though no wage variable appears in the equations. The value of labor power, a weight assigned to an hour performed, and that hour's actual duration are distinct; Section C.11 gives explicit counterexamples and primary sources. The equal-residual axiom is an additional model of the second object, not a proof that these objects coincide or that reproduction cost determines new value.

The additional spectral condition concerns solvability of this stationary closure. It does not establish cohort feasibility, social necessity, market validation, or a general Marxian law of skilled labor. Existing discussions of interpretation and limits in Sections 1 and 10 and Appendix B remain in force. The mathematical claim is conditional existence and uniqueness under the stated axiom and data. Boundary changes must preserve reproduction inputs as well as commodity requirements (Section C.9).

Complete specification. With fixed admissible data and $\rho(A_C) < 1$, $\rho(M_C) < 1$,

$$\begin{aligned}
G_C &= H_C(I - A_C)^{-1}, & R_C &= D_C \odot B_C^*, \\
M_C &= G_C S_C + (G_C E_C + U_C)\Gamma_C^{-1}, & & \\
\beta_C &= (I - M_C^{\mathsf{T}})^{-1}\mathbf{1}, & \alpha_C &= \beta_C/\beta_{C,b}, \\
v_C &= G_C^{\mathsf{T}}\alpha_C, & r_{ij}(C) &= \alpha_{C,i}/\alpha_{C,j}, \\
\ell_p(C) &= h_p + G_C a_p, & V_p(C) &= \alpha_C^{\mathsf{T}}\ell_p(C).
\end{aligned} \tag{C.19}$$

C.9 Exact reporting-boundary reduction for the conversion

Appendix A preserves the ordinary labor matrix when suppliers are omitted from a reporting boundary. It is not sufficient simply to delete the corresponding rows of S and E: workers can consume food, education, or services supplied outside that boundary. This subsection completes the boundary rule for the optional conversion model.

Partition the output nodes into retained L and omitted O, while retaining *all occupational coordinates*. With the block orientation of Appendix A, set

$$N_O = (I - A_{OO})^{-1}, \qquad A_{\text{eff}} = A_{LL} + A_{LO}N_O A_{OL}, \qquad H_{\text{eff}} = H_L + H_O N_O A_{OL}.$$

Write $F = \binom{F_L}{F_O}$, using $F = S + E\Gamma^{-1}$ and $T = U\Gamma^{-1}$.

Proposition C.8 (Conversion-preserving boundary elimination)**.** *For a productive full ordinary system, define*

$$\boxed{F_{\text{eff}} = F_L + A_{LO}N_O F_O, \qquad T_{\text{eff}} = T + H_O N_O F_O.} \tag{C.20}$$

Then, with $G_L = H_{\text{eff}}(I - A_{\text{eff}})^{-1}$,

$$\boxed{G_L F_{\text{eff}} + T_{\text{eff}} = GF + T = M.} \tag{C.21}$$

Consequently the admissibility condition $\rho(M) < 1$ *is unchanged. When it holds, the normalized weights and all labor-type ratios are unchanged, and the retained scalar values equal the corresponding entries of the full vector* v*.*

Proof. The external block of $G = H + GA$ gives

$$G_O = (H_O + G_L A_{LO})N_O.$$

Substituting into $M = G_L F_L + G_O F_O + T$ yields

$$M = G_L(F_L + A_{LO}N_O F_O) + (T + H_O N_O F_O),$$

which is (C.21). Appendix A supplies G_L. Since M and the reference type are unchanged, Theorem C.2 gives the same α, and $v_L = G_L^{\mathsf{T}}\alpha$. □

T_{eff} is an *effective boundary labor requirement*, not solely direct teaching labor. Likewise H_{eff} is not solely labor performed inside L. This reduction preserves a specified full context; it does not assert that different countries, techniques, or provisioning conventions have the same ratios. Deleting occupations requires additional aggregation assumptions and is not covered by this proposition.

Concrete failure of naive deletion. In the example of Sections C.6–C.7, omit the consumption-bundle and learning-material sectors from the reported outputs. The exact rule preserves M and $r_{\text{eng,prod}} = 11/8$. Simply deleting their rows from F instead leaves

$$M_{\text{naive}} = T = \begin{pmatrix} 0 & 0 \\ 3/100 & 1/10 \end{pmatrix}, \qquad r^{\text{naive}}_{\text{eng,prod}} = \frac{100}{93} \neq \frac{11}{8}.$$

That change is an accounting omission, not a new economic result.

C.10 Labor-unit covariance and the residual convention

The physical-unit invariance in Section C.5 does not permit an unequal rescaling of occupational time units while silently retaining equal numeric residuals. A residual per hour and a residual per eight-hour day are different quantities.

Let $J = \operatorname{diag}(t_1, \dots, t_m)$, with $t_i > 0$, where one new unit of type-i labor represents t_i old clock hours. Keep commodity units fixed and transform

$$\begin{aligned} H' &= J^{-1}H, & S' &= SJ, & E' &= E, \\ U' &= J^{-1}U, & \Gamma' &= J^{-1}\Gamma. \end{aligned} \tag{C.22}$$

The worker-count unit is unchanged; thus education goods per newly qualified worker remain E, while lifetime productive time is expressed in the new labor units.

Proposition C.9 (Covariance under occupational time units)**.** *With* (C.22),

$$G' = J^{-1}G, \qquad F' = FJ, \qquad T' = J^{-1}TJ, \qquad M' = J^{-1}MJ. \tag{C.23}$$

In the same common-value unit, let $\alpha' = J\alpha$ and $q' = J\mathbf{1}$. Then

$$(I - M'^{\mathsf{T}})\alpha' = \kappa q', \qquad G'^{\mathsf{T}}\alpha' = G^{\mathsf{T}}\alpha = v. \tag{C.24}$$

If the new reference weight is also reset to one, use

$$\widetilde{\alpha}' = \frac{J\alpha}{t_b}, \qquad \widetilde{\kappa}' = \frac{\kappa}{t_b}, \qquad \widetilde{v}' = \frac{v}{t_b}.$$

The latter change is an additional rebasing of the common-value unit.

Proof. The labor equation gives $G' = J^{-1}G$. Since J and Γ are diagonal, substituting (C.22) into F and T gives (C.23). Hence $M'^{\mathsf{T}} = JM^{\mathsf{T}}J^{-1}$ and

$$(I - M'^{\mathsf{T}})J\alpha = J(I - M^{\mathsf{T}})\alpha = \kappa J\mathbf{1}.$$

Also $G'^{\mathsf{T}}J\alpha = G^{\mathsf{T}}\alpha$. Finally $\alpha'_b = t_b\alpha_b = t_b$, proving the stated normalization. Similarity preserves $\rho(M)$. □

Example. Keep production labor in hours and express engineering in eight-hour days. For the original example, $J = \operatorname{diag}(1, 8)$ and $\alpha' = (1, 11)^{\mathsf{T}}$: one engineering day receives eleven reference-hour equivalents, exactly as eight hours at 11/8 each. The physical product, copy, and service values do not change.

Ratios stated *per new labor unit* transform as $r'_{ij} = (t_i/t_j)r_{ij}$ before any common rebasing. Ratios per original clock hour are recovered by dividing the relevant weights by t_i. Replacing $q' = J\mathbf{1}$ by $\mathbf{1}$ after an unequal rescaling would impose a different residual assumption, not describe the same system in different notation.

C.11 Interpretation: what is identified by the closure

The occupational requirement G_C is determined by the selected ordinary recipes. The pairwise conversion ratios are determined only after adding provisioning and training data *and* the equal-residual axiom. Calling the latter calculation endogenous is conditional on those choices; it is not an empirical identification of relative skill, marginal output, wage rates, or a universal law of abstract labor.

An explicit counterexample. Let $A = 0_{2\times 2}$, $H = I_2$, $E = U = 0$, and $S = \mathrm{diag}(1/5, 2/5)$. Both selected recipes require one clock hour for one output unit, and no difference in qualification cost is specified. Nevertheless,

$$G = I_2, \qquad M = \begin{pmatrix} 1/5 & 0 \\ 0 & 2/5 \end{pmatrix}, \qquad \beta = \begin{pmatrix} 5/4 \\ 5/3 \end{pmatrix}, \qquad \alpha = \begin{pmatrix} 1 \\ 4/3 \end{pmatrix}.$$

The unequal weights result from the unequal declared provisioning accounts under the axiom. They do not demonstrate that the second worker is more skilled or technologically more productive. This distribution sensitivity is a consequence of the chosen model, not an algebra error.

Likewise, for the original example's *same* M, replacing the residual $\kappa\mathbf{1}$ by $\kappa(1, 2)^{\mathsf{T}}$ changes $r_{\mathrm{eng,prod}}$ from 11/8 to 38/17. Therefore the equal-residual assumption must accompany any reported ratio; it cannot be inferred from the numerical matrix alone.

Economic distinctions. Marx treats the social reduction of skilled labor in Volume I, Chapter 1, Section 2 [C1]; subsistence and training enter his discussion of the value of labor-power in Chapter 6 [C2]. Chapter 7 distinguishes the value of labor-power from the value produced through its exercise [C3]. These passages motivate relevant distinctions but do not establish the equal-residual equation used here. In particular, a provisioning burden, a weight applied to performed labor, and actual labor duration must not be substituted for one another.

Accounting consequence. $\alpha^{\mathsf{T}} G y$ is a scalarization of an occupational account, not an extra activity. A separate plan that actually produces food or supplies training still has to record that work once. Conversely, recursively adding M-generated rounds to a ledger whose corresponding work has already been counted would double count. If c_j^{rep} is used as the value of labor-power, the relation $\alpha_j - c_j^{\mathrm{rep}} = \kappa$ is *imposed* by this closure, not independently measured surplus value or an inferred exploitation rate.

Primary-source references for this clarification

[C1] Karl Marx. *Capital*, Volume I (1867), Chapter 1, Section 2. English transcription, Marxists Internet Archive.

[C2] Karl Marx. *Capital*, Volume I (1867), Chapter 6. The Buying and Selling of Labour-Power.

[C3] Karl Marx. *Capital*, Volume I (1867), Chapter 7, Section 2. The Labour-Process and the Process of Producing Surplus-Value.

C.12 Audit record and reproducible checks

Audit outcome. No arithmetic or algebraic change was required to the three core production/capability/project equations, the equal-residual solution, the augmented block inverse, or the original worked examples. The proofs remain conditional on their stated assumptions. The review does not establish empirical validity or constitute independent peer review.

The revised edition makes the following corrections and completions:

1. It supplies the missing extension of reporting-boundary elimination to workers' provisioning and qualification inputs (Section C.9), including a counterexample to naive deletion.
2. It gives the full labor-unit transformation, including the residual source vector and normalization (Section C.10). This completes the earlier prose warning.
3. It states the identification limitation concretely through two exact counterexamples and primary-source distinctions (Section C.11). The optional weighting model is retained, not silently replaced.
4. It replaces the embedded verification script's environment-specific output path. The previous script completed its assertions but failed when its hard-coded output directory was absent. The replacement prints to standard output by default and accepts an optional output path, creating parent directories when requested.

Verification bundle. The revised PDF embeds the editable appendix source, `verify_model.py`, and `verification.json`. The script uses Python 3 and SymPy, with no network access or empirical data. It was run successfully with

```
python verify_model.py -output verification.json
```

There are 135 named checks. These include the original physical/software examples and conversion example; exact symbolic derivatives; block inverse and scalar identities; physical-unit and labor-unit transformations; boundary elimination; and counterexamples for missing inputs and inadmissible reproduction matrices. Twenty seeded rational systems exercise cyclic supply chains and boundary feedback. One named check exhausts all 64 two-node Boolean graph/label configurations and verifies leastness, not only a fixed-point residual. Tests supplement the analytic proofs; a finite suite cannot prove their universal statements.

Numerical reliability. For a productive M, let $W = (I - M^{\mathsf{T}})^{-1} \geq 0$. If an approximate solution is $\widehat{\beta}$ and its exact residual is $r = \mathbf{1} - (I - M^{\mathsf{T}})\widehat{\beta}$, then

$$\beta - \widehat{\beta} = Wr, \qquad |\beta - \widehat{\beta}| \leq W|r|.$$

This follows by subtracting the linear systems and using nonnegativity of W. Near singularity, a small residual alone does not imply a small forward error. Certified bounds require verified input enclosures and arithmetic, or an independently justified inverse bound. If bounds $0 < l_i \leq \beta_i \leq u_i$ are available, they yield

$$\frac{l_i}{u_j} \leq r_{ij} \leq \frac{u_i}{l_j}.$$

The exact rational checks avoid floating-point uncertainty in the stated examples, but not uncertainty in future empirical input data.